FIRST ASSIGNMENT – HIGH BANDWIDTH GEO SATELLITE

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WHAT ARE GEOSTATIONARY SATELLITES

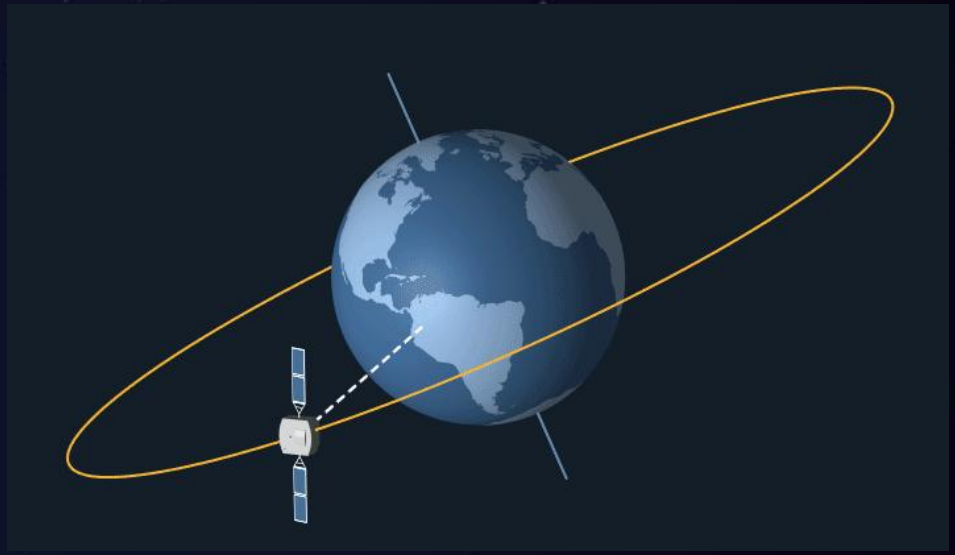
- Always above the same place on earth
- Very specific orbit

Altitude: 35,768 Km above sea level

inclination : 0 degrees (Above the equator)

Eccentricity : 0 (orbit is a perfect circle)

Velocity: 3.07 Km/s



Due to the above, a GEO orbit is only defined by the longitude (x degrees East or West)

EXAMPLES RESEARCHED

- Eutelsat KONNECT VHTS -> Ka - band, 500Gbps, DVB - S2X
- Eutelsat 10B -> C – band, Ku – band, 35Gbps
- Eutelsat KONNECT -> Ka – band, 75Gbps, DVB - S2X

- Most GEO high throughput satellites use multiple frequency bands or even multiple frequencies within the same bands, as shown before. We won't consider such approach. We will use one specific carrier frequency.

MULTIPLE TRANSPONDERS

- High throughput satellites use multiple beams in the same frequency to different geographic locations (frequency reusing). This means that they can send 5Gbps to Athens and 5Gbps to Paris in the same frequency, with minimal interference. For example:

Eutelsat 10B has 36 Ku – band transponders [1]

Eutelsat connect has 65 spotbeams [2]

- If we had 10 transponders the bandwidth we would need would be 1/10 of what we need now

We analyzed the feasibility for a single transponder, but the bandwidth required was too big (>1GHz). The economic needed to buy such a large bandwidth, whatever the frequency band, is prohibiting

ASSUMPTIONS

- Satellite can only use 1 continuous spectrum
- We send and receive data to 10 station
- Same frequency for downlink and uplink, can't do both simultaneously
- Filter: raised cosine with a = 0.2
- Rb = 10Gbps or 1Gbps per transponder
- Bandwidth BW = $\frac{Rb}{10 * k} * (1 + a) = \frac{1.2Gbps}{k}$, k being the modulation order (bits / symbol)

MODULATION

1st choice: 256 - APSK

- High speed
- Lowest bandwidth usage(Modulation order = 8)
- Needs good weather, very susceptible to noise

Bandwidth = 150MHz

2nd choice: 64 - APSK

- Slower
- Modulation order = 6
- Middle point

Bandwidth = 200MHz

3nd choice: Low order PSK (8 - PSK)

- slowest
- Modulation order = 3
- Good inter – symbol distance
- Less affected by noise

Bandwidth = 400MHz

* All the above are used by the DVB – S2X protocol

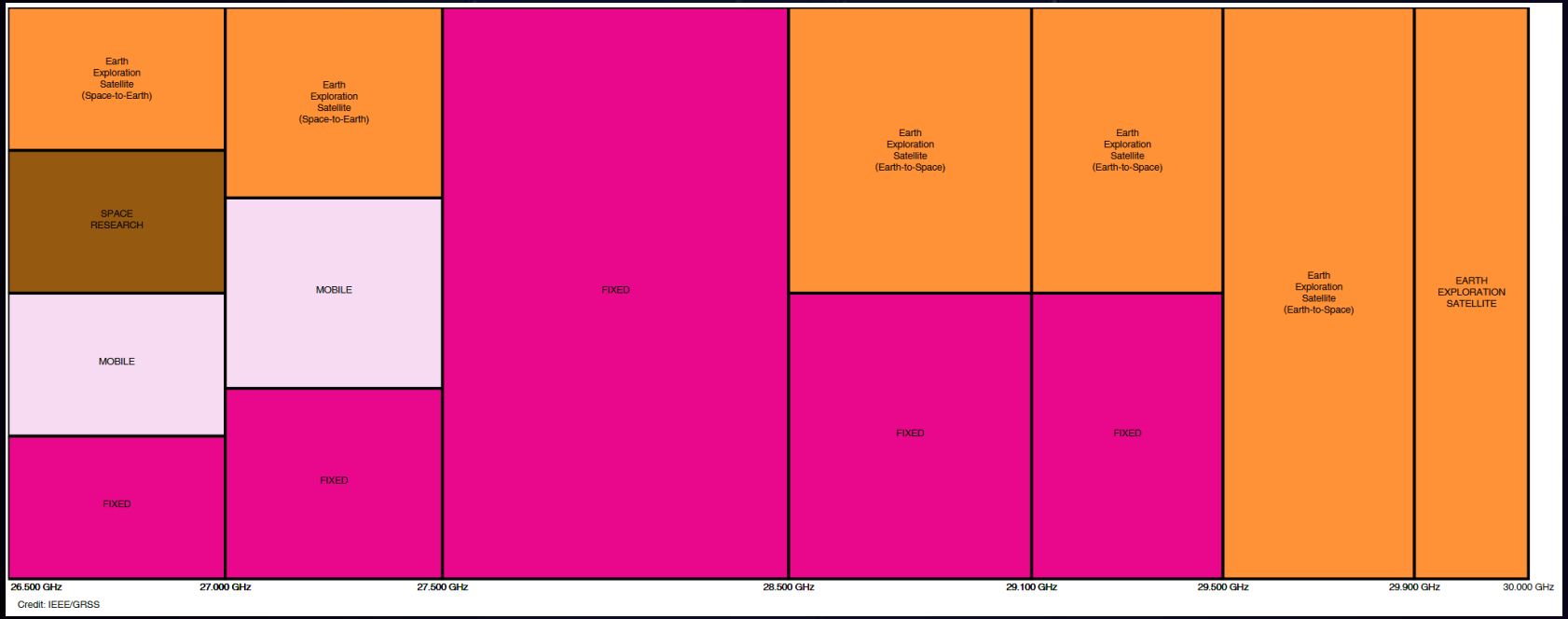
FREQUENCIES

Candidate frequencies

- High throughput satellites often use the Ka-bands. Examples:
- Other candidate Ku-band
- Q/V band
- Radio amateur

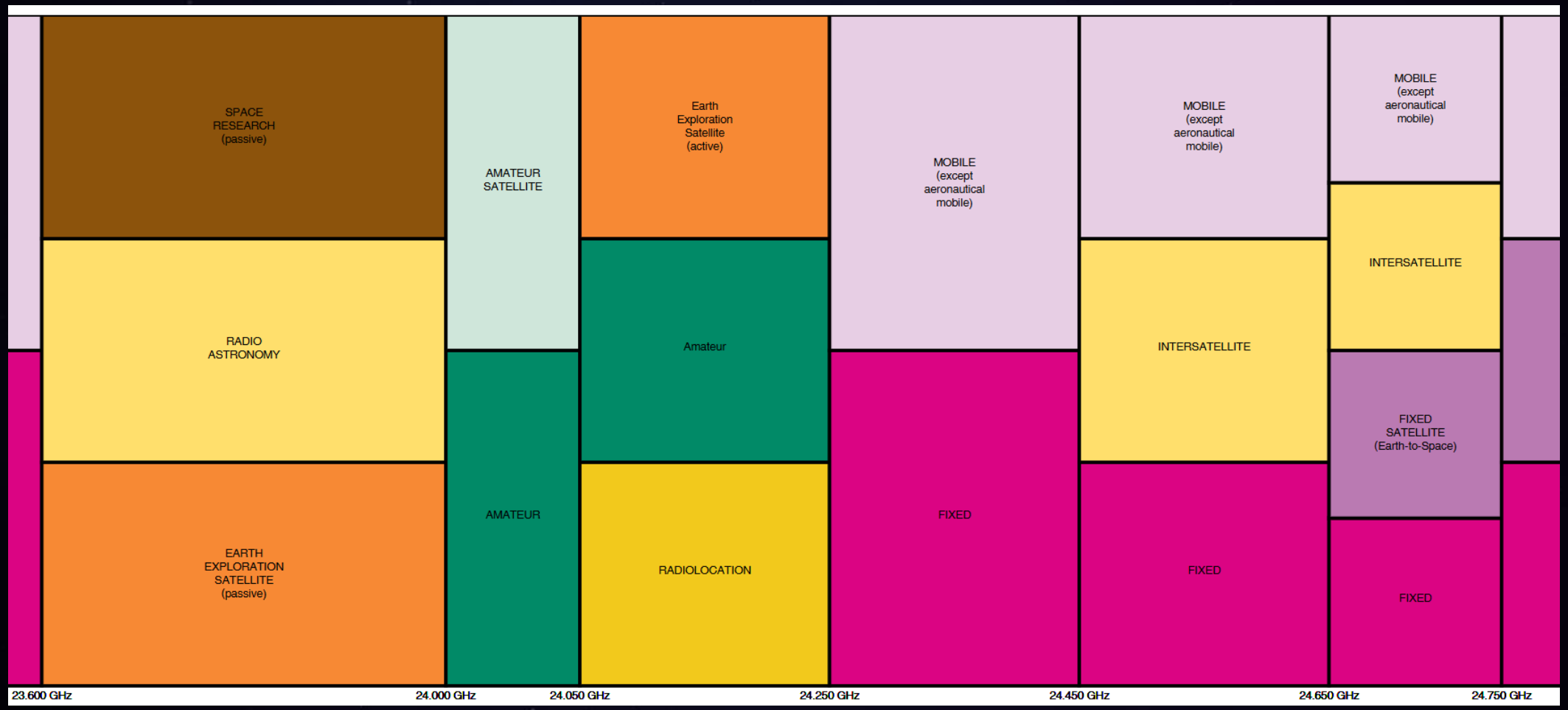
KA BAND

Uplink 27.5 – 30 GHz, Downlink 17.7 - 20.2 GHz



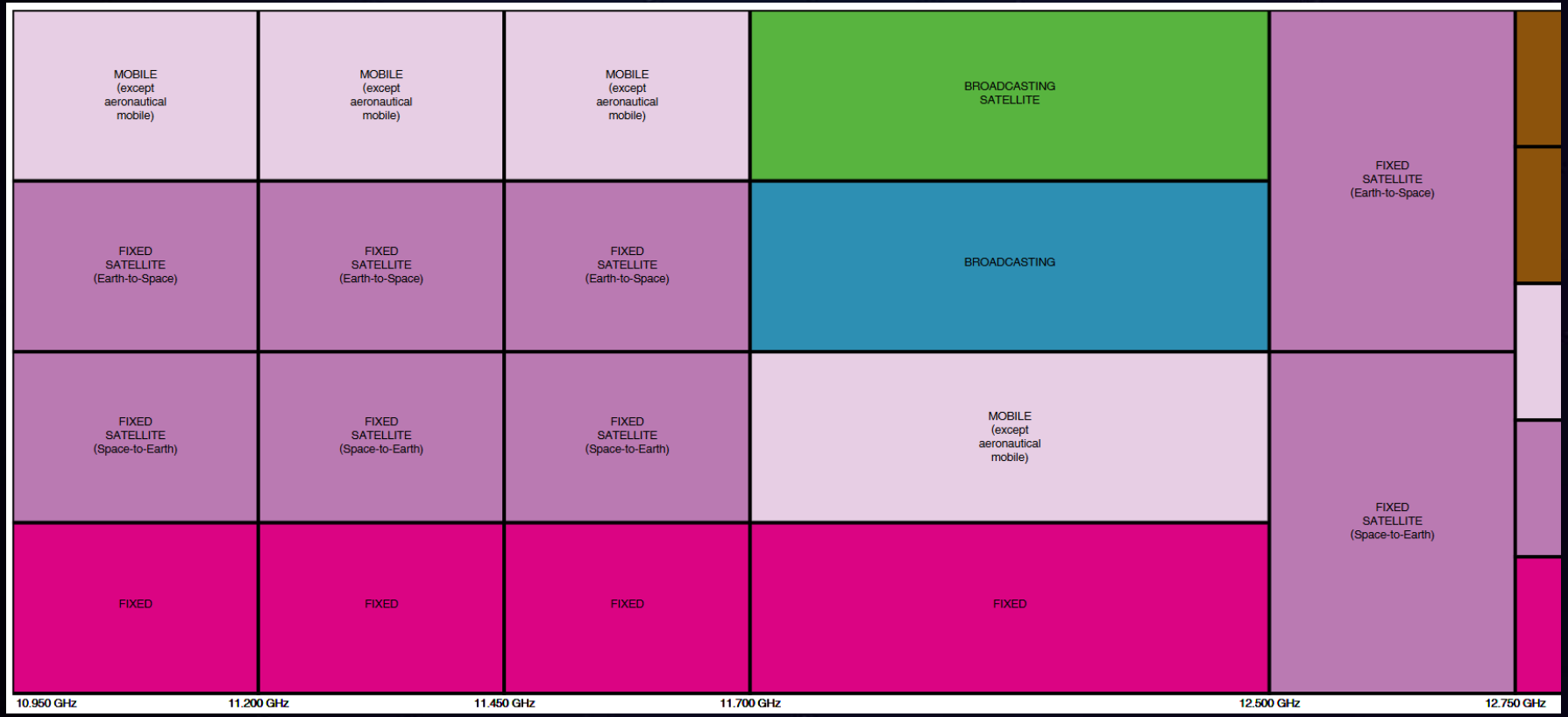
https://tools.grss-ieee.org/fars_allocations/frequency_allocations.html

KA BAND (AMATEUR)



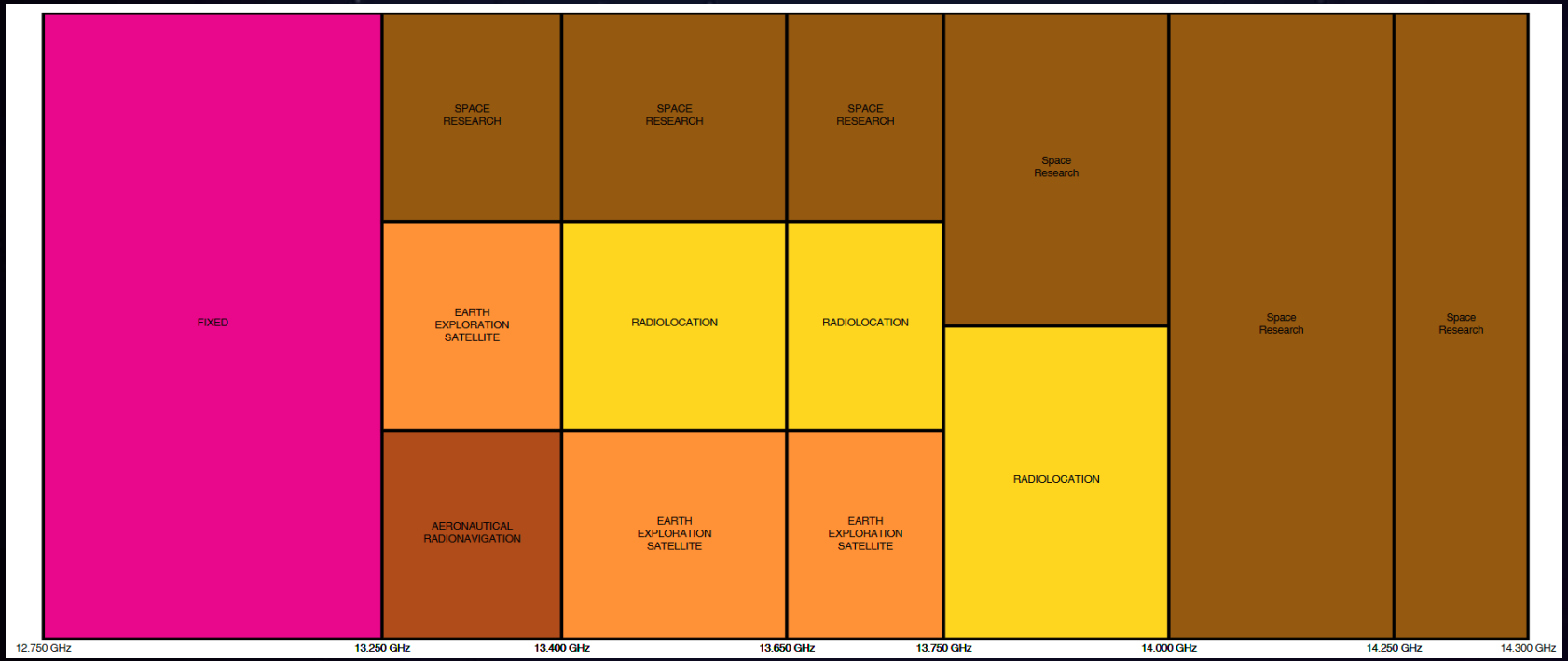
KU BAND

Uplink 14 – 14.5 GHz, Downlink 11.5 – 12.2 GHz



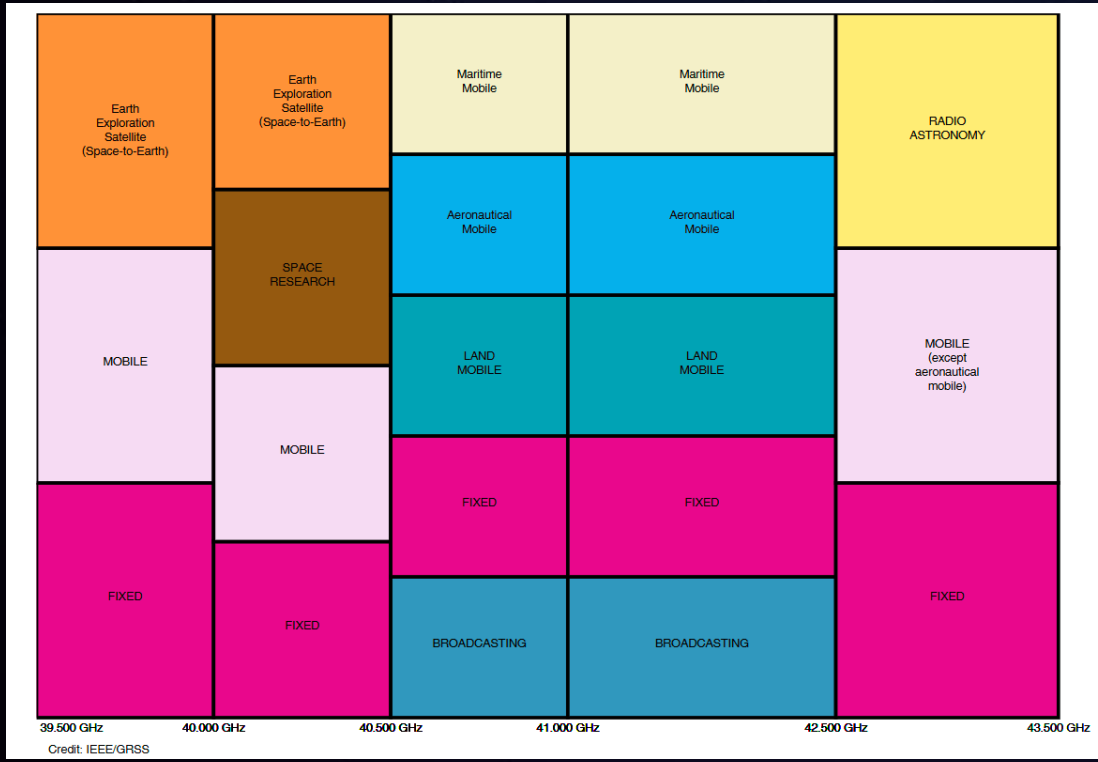
KU BAND

Uplink 14 – 14.5 GHz, Downlink 11.5 – 12.2 GHz



Q/V BAND

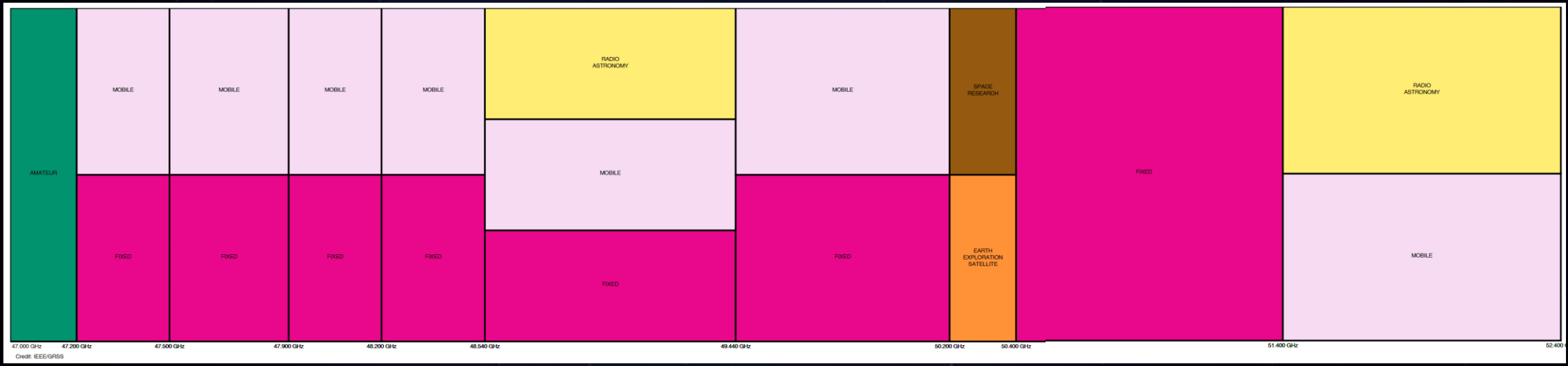
Uplink 47.2 – 51.4 GHz, Downlink 37.5 – 42.5 GHz



Q band

Q/V BAND

Uplink 47.2 – 51.4 GHz, Downlink 37.5 – 42.5 GHz



V band

EXAMPLE OF KU BAND

Identity

Satellite Network Name:	EUTELSAT 1-8
Satellite Network Previous Name:	EUTELSAT 1-6 EUTELSAT 1-8
Administration:	F (France)
Network Organization:	EUT (EUTELSAT)
Space Plan/Non-Plan Bands:	Non-Plan
Type of Satellite Network:	Geostationary (25.5° East)

-180 0 180

Regulatory Status Tracking

Status:	Active
In Master Register (MIFR):	Yes
Period of Validity:	75 years
Confirmed Brought into Use:	Yes (Full)
Suspension of Use:	No
Earliest Regulatory Date Limit of Bringing into Use:	12.09.1998

Note: Data displayed is the latest available of the Related Notices Information List.

Publications in BR IFIC Space

4 API	**10** COORDINATION
9 NOTIFICATION	**1** OTHER

Earliest Publication Date: 15.02.1994
Latest Publication Date: 27.11.2018

Related Notices Information

Notice ID	Provision	Notice Date of Receipt
Notification		
▶ 101500027	11.2	29.11.2016
Other Special Section		
98590455	RS49	30.09.1998

Network Structure Navigation

	Lower	Frequency Limits	Upper
Emission (Tx)	10'955.67 MHz	(8 bands)	11'694.33 MHz
Reception (Rx)	14'005.67 MHz	(6 bands)	14'494.33 MHz
Inter-Satellite Link (ISL)	MHz	(0 bands)	MHz

https://www.itu.int/itu-r/space/apps/public/spaceexplorer/networks-explorer/space-stations/dashboard/non-plans/EUTELSAT%201-8/101500027/Identity

EXAMPLE OF KA BAND

Identity

Satellite Network Name:	EUTELSAT-KA-36E
Satellite Network Previous Name:	—
Administration:	F (France)
Network Organization:	EUT (EUTELSAT)
Space Plan/Non-Plan Bands:	Non-Plan
Type of Satellite Network:	Geostationary (36° East)

-180 0 180

Regulatory Status Tracking

Status:	Active
In Master Register (MIFR):	Yes
Period of Validity:	40 years
Confirmed Brought into Use:	Yes (Full)
Suspension of Use:	No
Earliest Regulatory Date Limit of Bringing into Use:	19.03.2005

Note: Data displayed is the latest available of the Related Notices Information List.

Publications in BR IFIC Space

2 API	**7** COORDINATION
6 NOTIFICATION	**3** OTHER

Earliest Publication Date: 19.03.1996
Latest Publication Date: 13.12.2011

Related Notices Information

Notice ID	Provision	Notice Date of Receipt
Notification		
▶ 108500578	11.2	26.07.2011
Other Special Section		
104591532	RS49	23.03.2011

Network Structure Navigation

	Lower	Frequency Limits	Upper
Emission (Tx)	17'700 MHz	(2 bands)	20'200 MHz
Reception (Rx)	27'500 MHz	(2 bands)	30'000 MHz
Inter-Satellite Link (ISL)	MHz	(0 bands)	MHz

https://www.itu.int/itu-r/space/apps/public/spaceexplorer/networks-explorer/space-stations/dashboard/non-plans/EUTELSAT-KA-36E/108500578

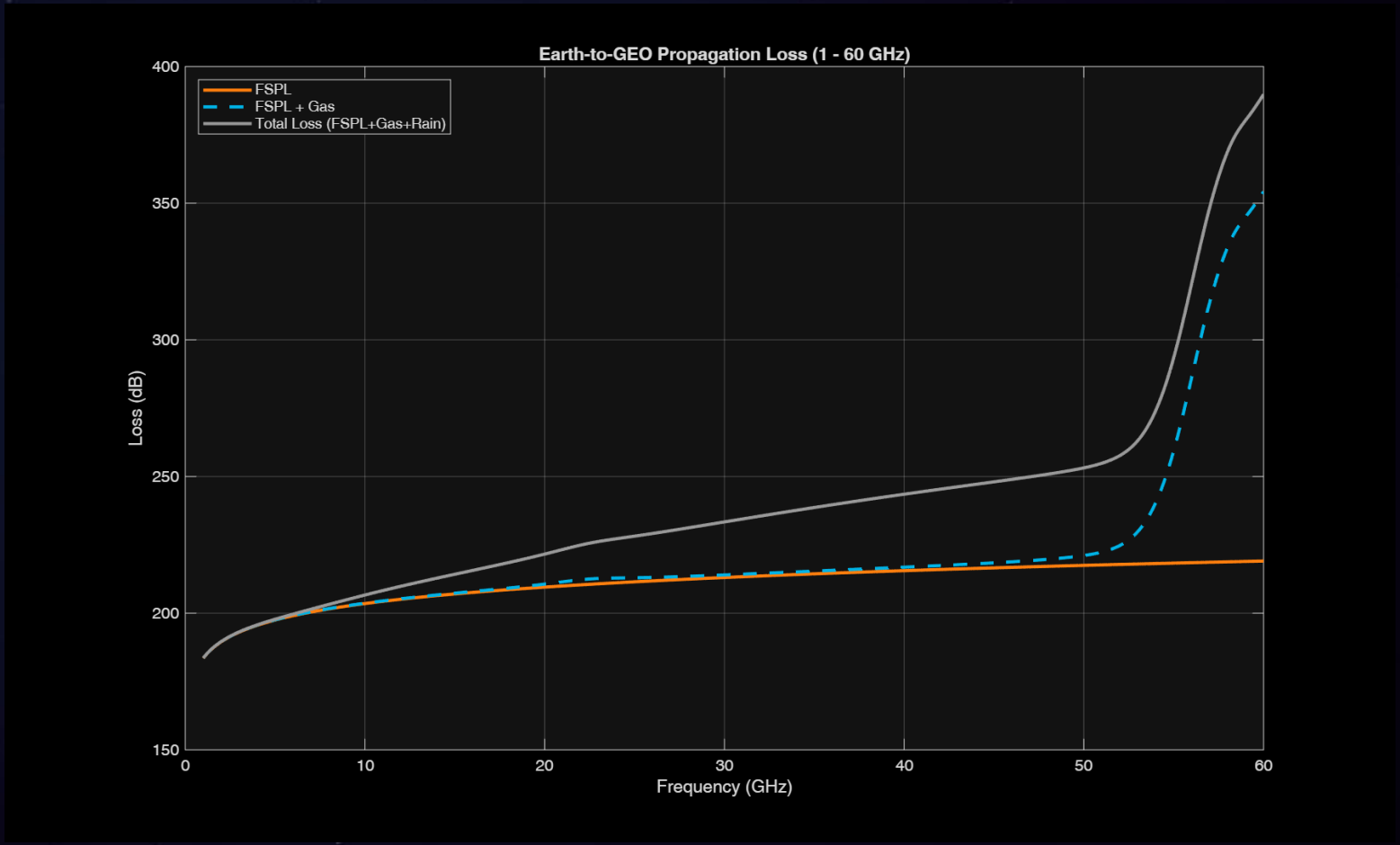
SIGNAL LOSS CALCULATIONS

- Rain set to 45mm/h.

Worst-case scenario,

weather paper, page 5 (275)

- Water vapor at 7.5 g/m^3



FREQUENCY

1st choice: Ku – band

- good speed
- fairly populated and expensive
- Minimal losses due to weather
- Larger antennas needed, more weight

We chose the 11.2GHz due to being slightly below what most high throughput GEO satellites use, resulting in lower cost

2nd choice: Ka – band

- High speed, big bandwidths
- Less populated
- Affected by weather
- Smaller antennas

We chose the 30GHz due to being low utilization, even though we will need more power due to the weather having a bigger impact on losses

3rd choice: radio amateur Ka band

- Same as Ka band
- FREE !!!
- Very limited bandwidth

24 - 24.25GHz

We chose the 24.125GHz, the center of the band

Rejected: Q/V band

Rejected due to high losses from weather. Since we only have 1 frequency, this will make communication almost impossible when in bad weather

FINAL CHOICES

1st choice: Ku – band

Carrier at 11.2GHz

256 – APSK

Bandwidth needed: 150MHz

Reason: low losses due to weather so we can use high order modulations. Ku band is congested so we need low bandwidth to reduce cost

2nd choice: Ka – band

Carrier at 30GHz

8 - PSK

Bandwidth needed: 400MHz

Reason: high weather losses force us to use a lower order modulation for signal integrity. Additionally, the band isn't very congested, and we can buy higher bandwidth

3rd choice: radio amateur Ka band

Carrier at 24.125GHz

64 - APSK

Bandwidth needed: 200MHz

Reason: max bandwidth for good signal integrity. Lowest cost option. We might need to invest in better filters to reduce the bandwidth.

COST CALCULATION

We used the SpaceCap v10 software to calculate the approximate cost of our bands

For Ka band we selected the EUTELSAT-KA-16E network for a quick cost calculation, with 1 beam

For Ku band we selected the EUTELSAT-KU-25.5E network, with 1 beam

Cost was the same for both at 19,234 CHF

Not all parameters were filled as some where highly technical and outside the scope of this search. This might have led to miscalculations.

KA BAND COST CALCULATION

Notice Id: 6 | AP4/II and AP4/III (Appendix 4 - Annex 2A) | 03/14/2026 | Status 01

Notice submitted under/for:

- No. 9.6 Coordination
- No. 11.2 Notification — First Notification
- Subj. to Coord. Sect.II Art.9 — 11.32 / 11.32 + 11.32A / Not Subj. to Coord. Sect.II Art.9
- No. 9.11A Applies
- Bands 21.4 to 22 GHz Special Procedure
- Specific Earth Station Coordination under No. 9.7A
- Earth Station Coordination under No. 9.21
- Earth Station Coordination between Administrations under No. 9.17

Date: MM/DD/YY 03/14/2026 | Administration Serial Nbr AUTH1

A1f1. Notifying Administration: GRC

A1f2. Notice submitted on behalf of these administrations.

A1f3. Intergovernmental Satellite System

Notice intended for: Addition / Modification / Suppression

BR Identification No. of Station to be modified/suppressed

Type of Satellite Network or Earth Station: GeoStationary Satellite Network / NonGeoStationary Satellite Network

A5./A6. Coordination Agreements: Group Level Agreement

A5a2a./A6a1. Coordination Agreements: Satellite Network List

Notice Id: 6 | Administration: GRC | Satellite Network: EUTELSAT-KA-16E

Notice 6 | Satellite Network: EUTELSAT-KA-16E | Beam Id BEAM1 E | Group Id: 1

3. Observed Frequencies and Related Characteristics: Add / Mod / Sup of the group | BR Identification of the Group to be modified/suppressed | Page No.

Characteristics Common to a Group of Frequencies | General Characteristics

C3a. Assigned Frequency Bandwidth: 400000 (kHz)

No Sensors / Active Sensors / Passive Sensors

C4a. Cls Stn	C4b. Nat Srv

C6. Polarization: Type M Mixed Polarization | If linear, provide angle °

C2c. Frequency assignments are filed under No.4.4

BR97 No.11.43A

BR98 For use in accordance with Resolution 163/164

C8d1. Maximum Total Peak Power: dBW

C8d2. Contiguous Bandwidth: (kHz)

C11a. Service Area (to be provided in Gims)

Service Area No. (diag provided in Gims)

Remarks

Characteristics of the Beam

B2. Receiving Beam / Transmitting Beam

B1a. Beam Designation: BEAM1 | Old Beam Designation (if changed)

B1b. Steerable Beam | Add / Mod / Sup of the Beam | Beam has Sensors

Antenna Characteristics

B3a1. Maximum Isotropic Gain +/- dBi: 60

B3d. Pointing Accuracy Degrees +/-: 3

Antenna Radiation Pattern

B3c1. Co-polar Radiation Pattern Id:

or B3c1 Pattern in the form of equations/diag. See Attach no.

List of Available Groups

Group 1 Page No.

A1a. Identity of the Satellite Network: EUTELSAT-KA-16E | A1c.

A4a. For GeoStationary Satellites Only

1. Nominal Orbital Longitude: Degrees 16.00 | E/W E

2. Longitudinal tolerance (degrees): a. To West 2 | b. To East 2

2c. Inclination Excursion: °

A.17b,d,e,f Compliance with PFD limits: Enter PFD values

Commitments: Manage Commitments

(Please use this function only after all frequency assignments have been captured in the notice)

List of Available Beams

Beam BEAM1

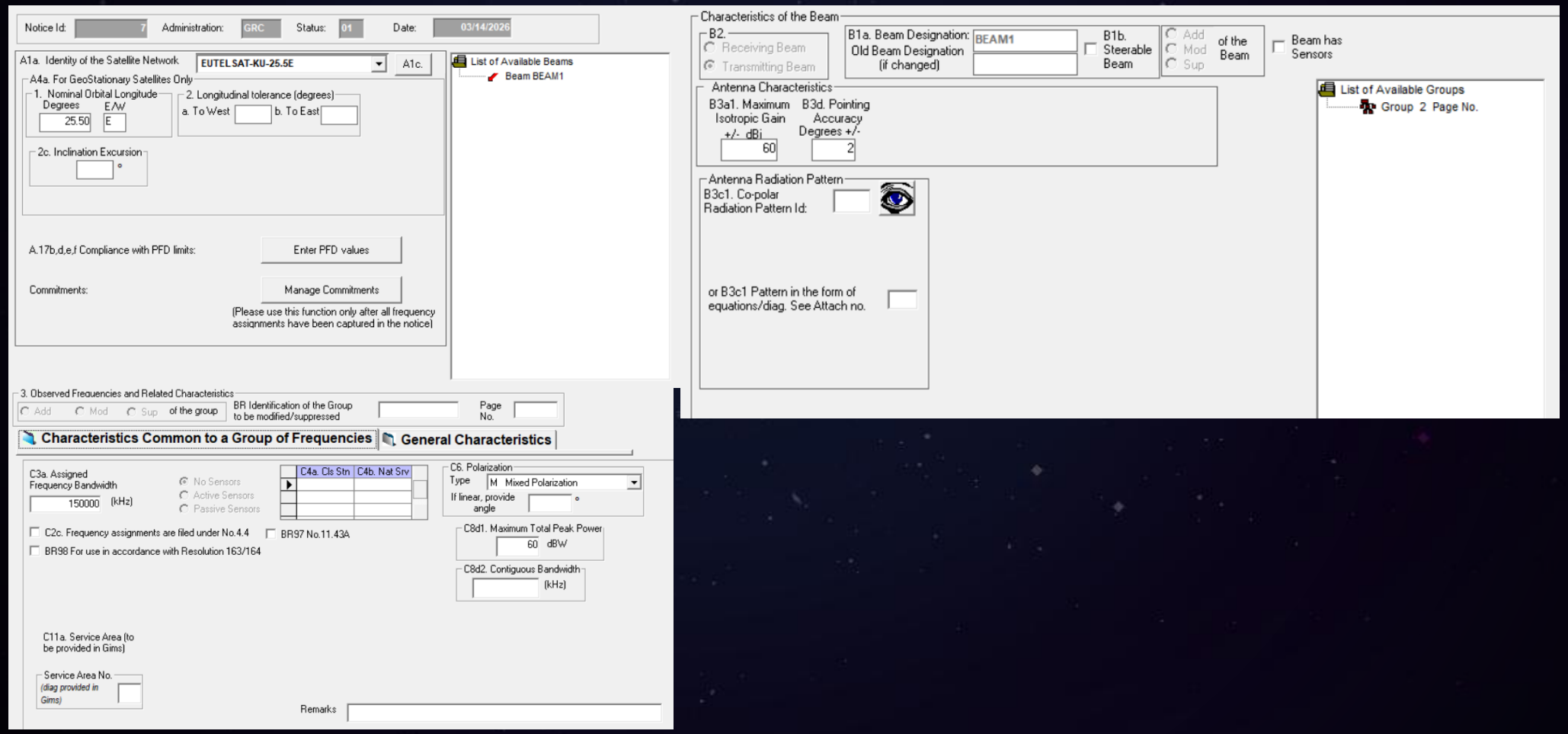
```
NOTICE: 6 EUTELSAT-KA-16E - Total number of unique groups processed: 1

For networks received from January 1, 2026 all fees must be multiplied by 1.1006 (fee multiplier)

 - Provn     :
 - Cat       : N4
 - Unit      : 0
 - FlatFee   : 12300
 - AmountDue: 6933.78      * Fee Multiplier applied: 1.1006


 - GROUP: 1 [E], freq:[0-0]
       - Provn          :
       - StnCls         :
       - BeamName       : BEAM1
       - BeamActCode    :
       - GroupActCode   :
```

KU BAND COST CALCULATION



CALCULATIONS FOR MATLAB

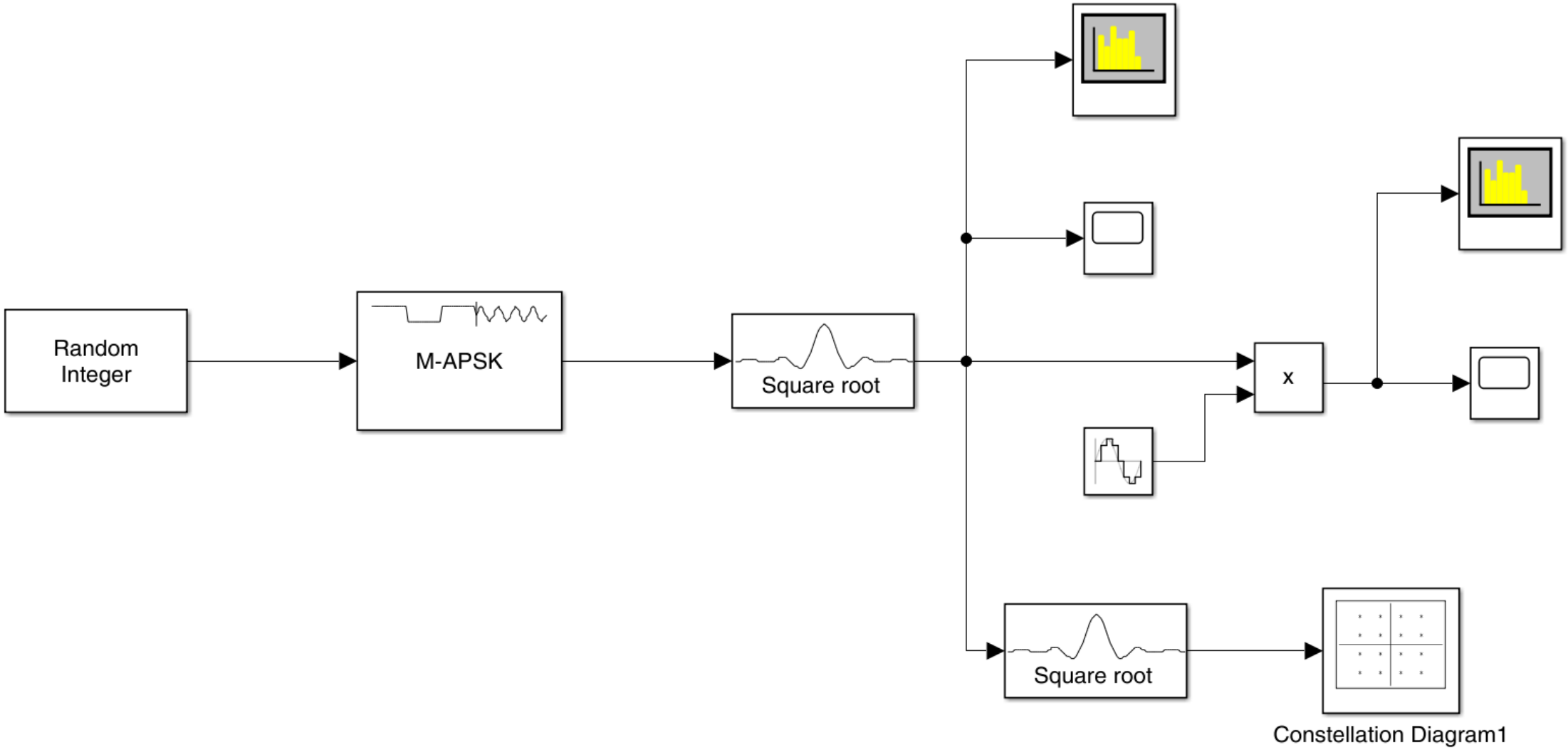
Random integer generator -> [256, 64, 8] for [256 – APSK, 64 – APSK, 8 - PSK]

Sample time = $\frac{1}{\frac{\text{Gbps}}{\text{k}}} = \frac{k}{Gbps} = \frac{k}{1e9}$ = [8E-9, 6E-9, 3E-9]

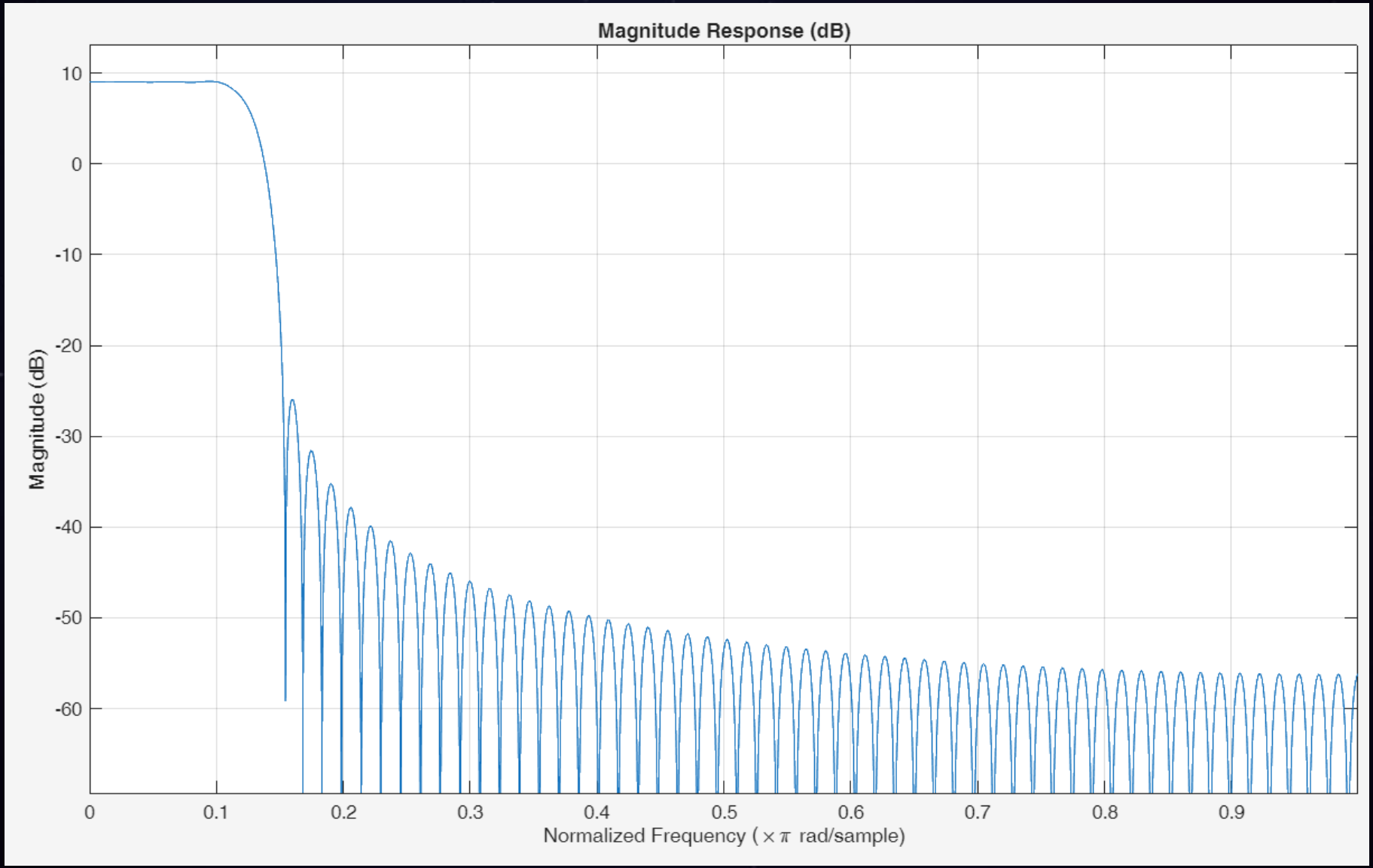
according to DVB – S2X protocol, 256 – APSK on page 37, 64 – APSK on page 32 and 8 – PSK on page 27

Raised cosine filter with a = 0.2 (conservative)

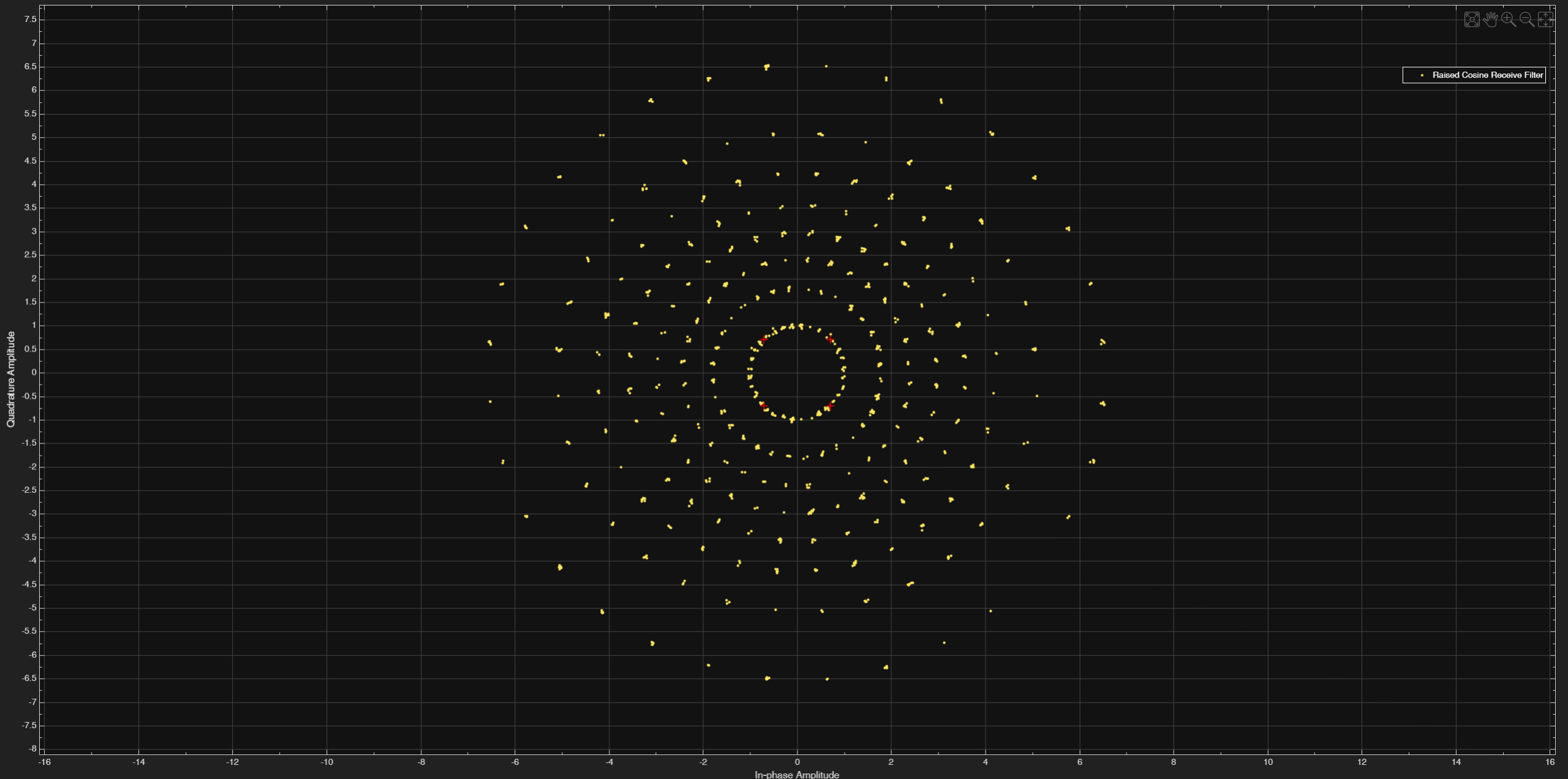
MATLAB BLOCK DIAGRAM



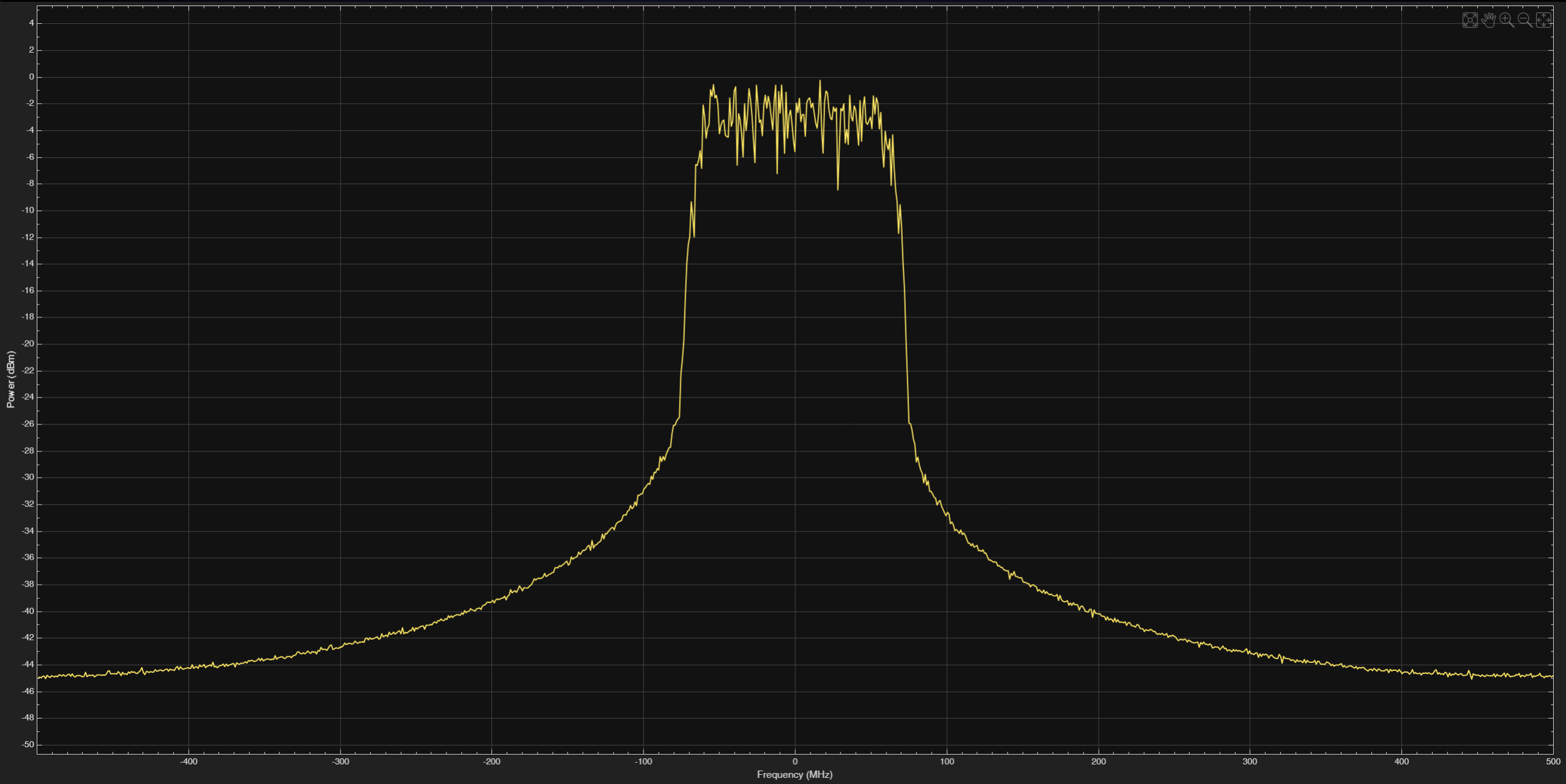
RAISED COSINE FILTER RESPONSE



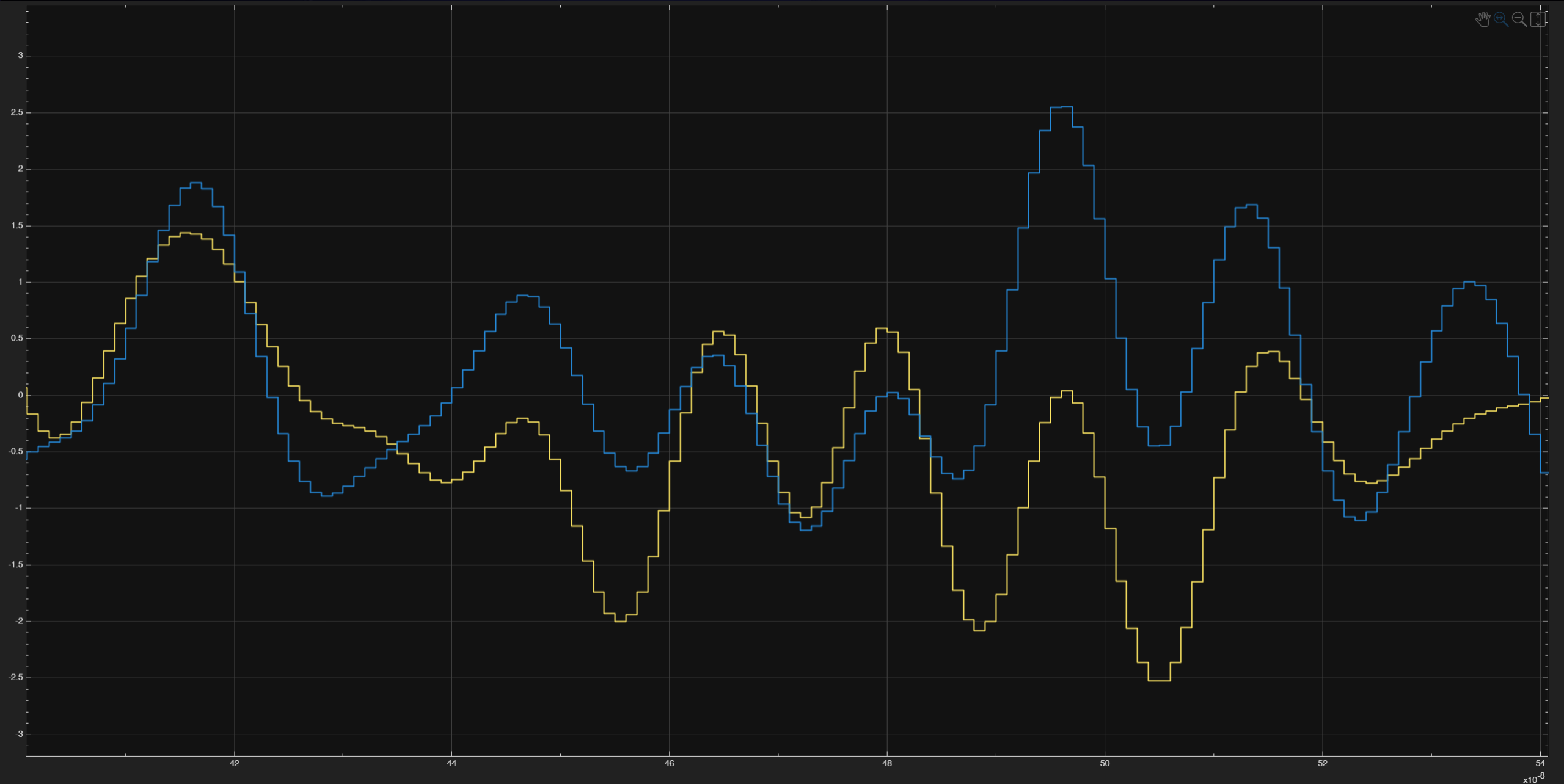
256 – APSK CONSTELLATION



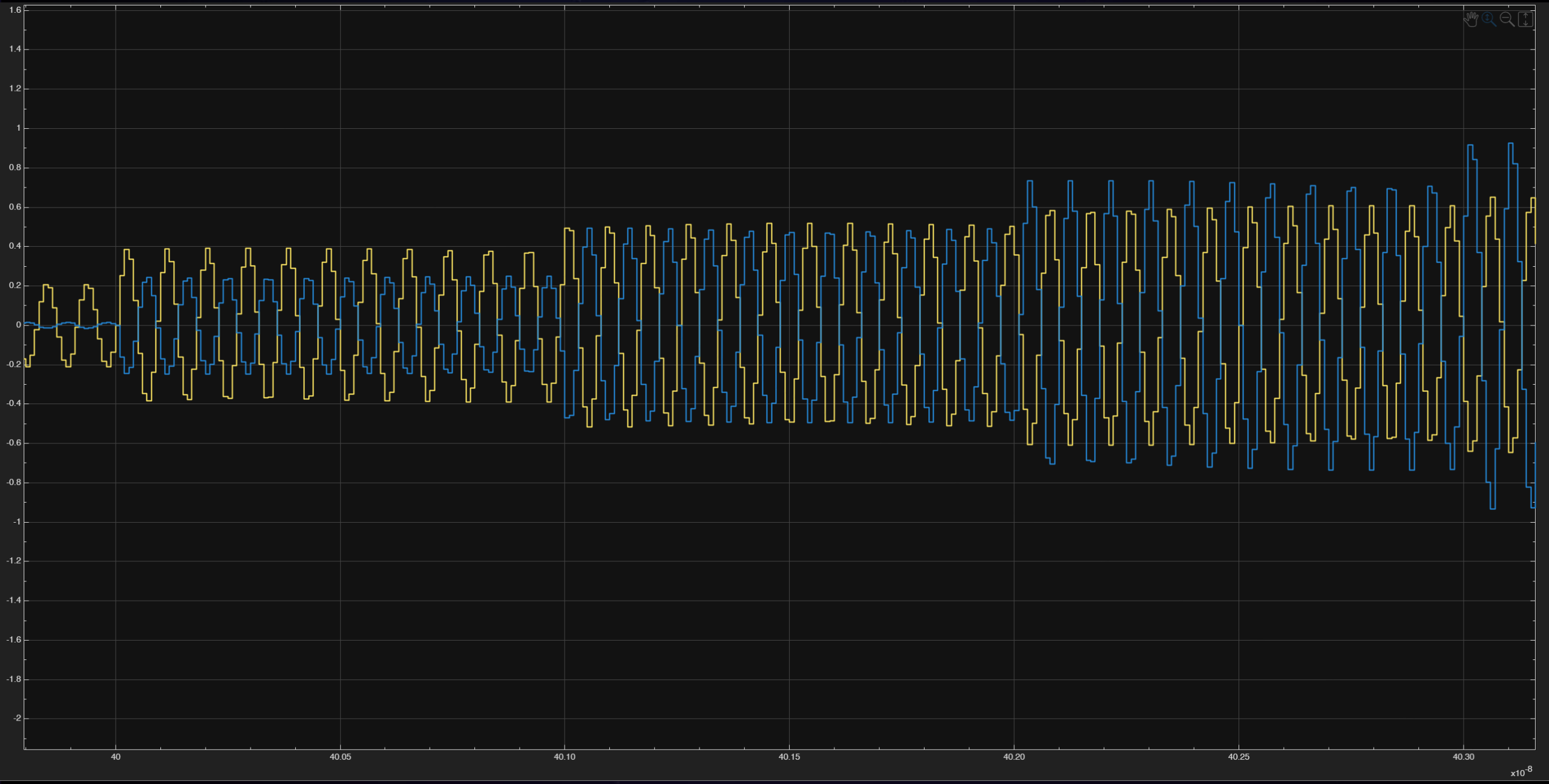
256 – APSK SPECTRUM



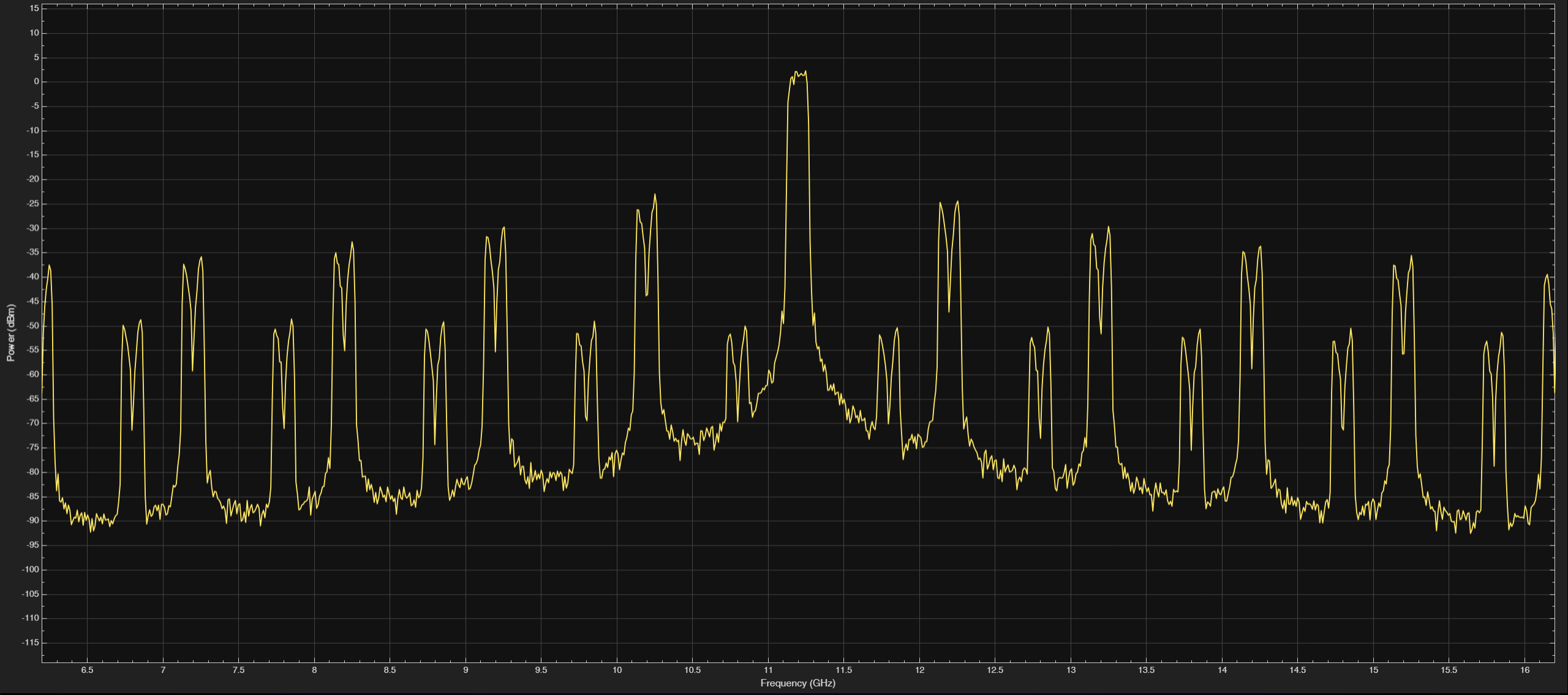
256 – APSK TIME DOMAIN



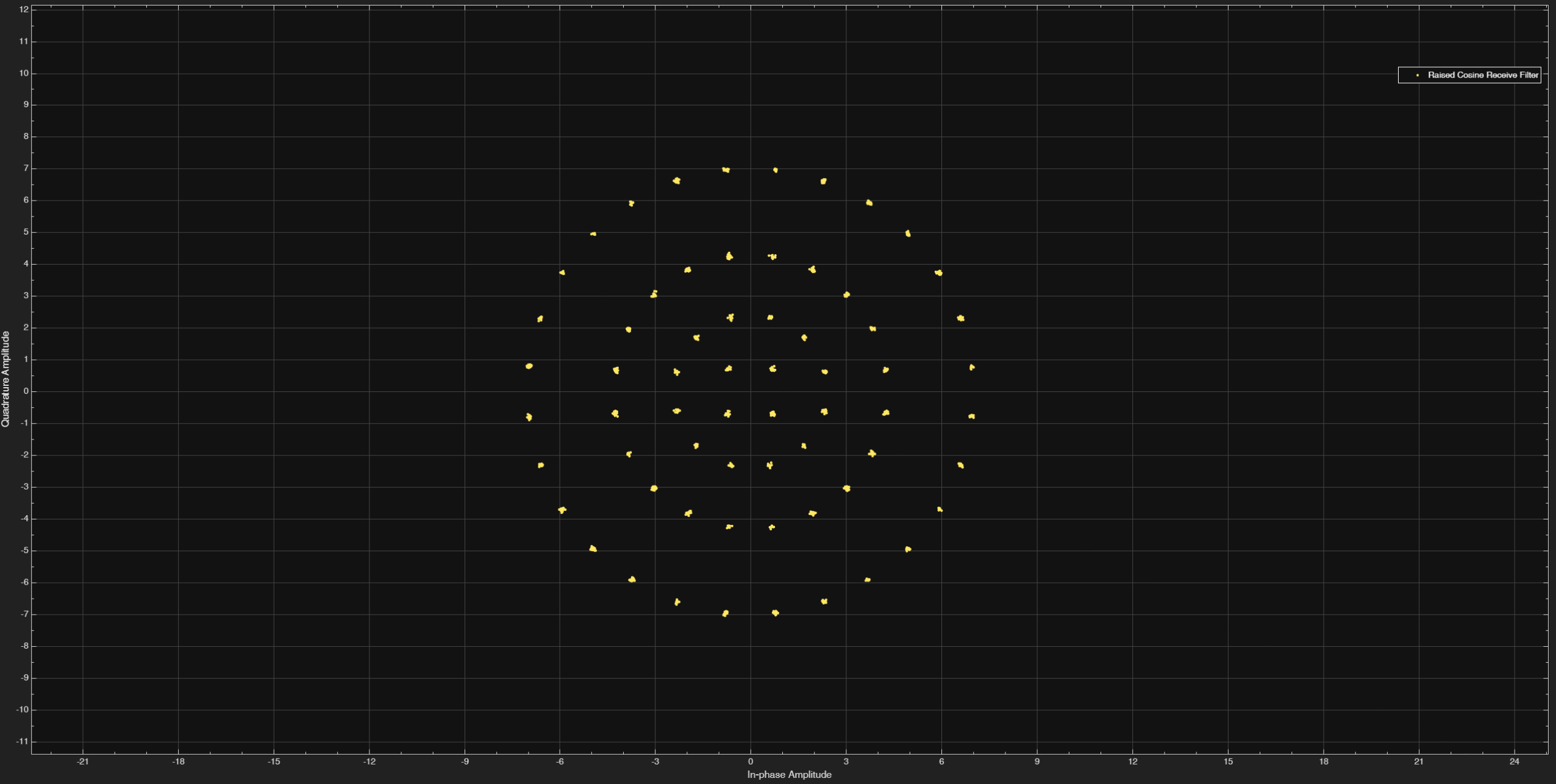
256 – APSK TIME DOMAIN AFTER UP CONVERSION



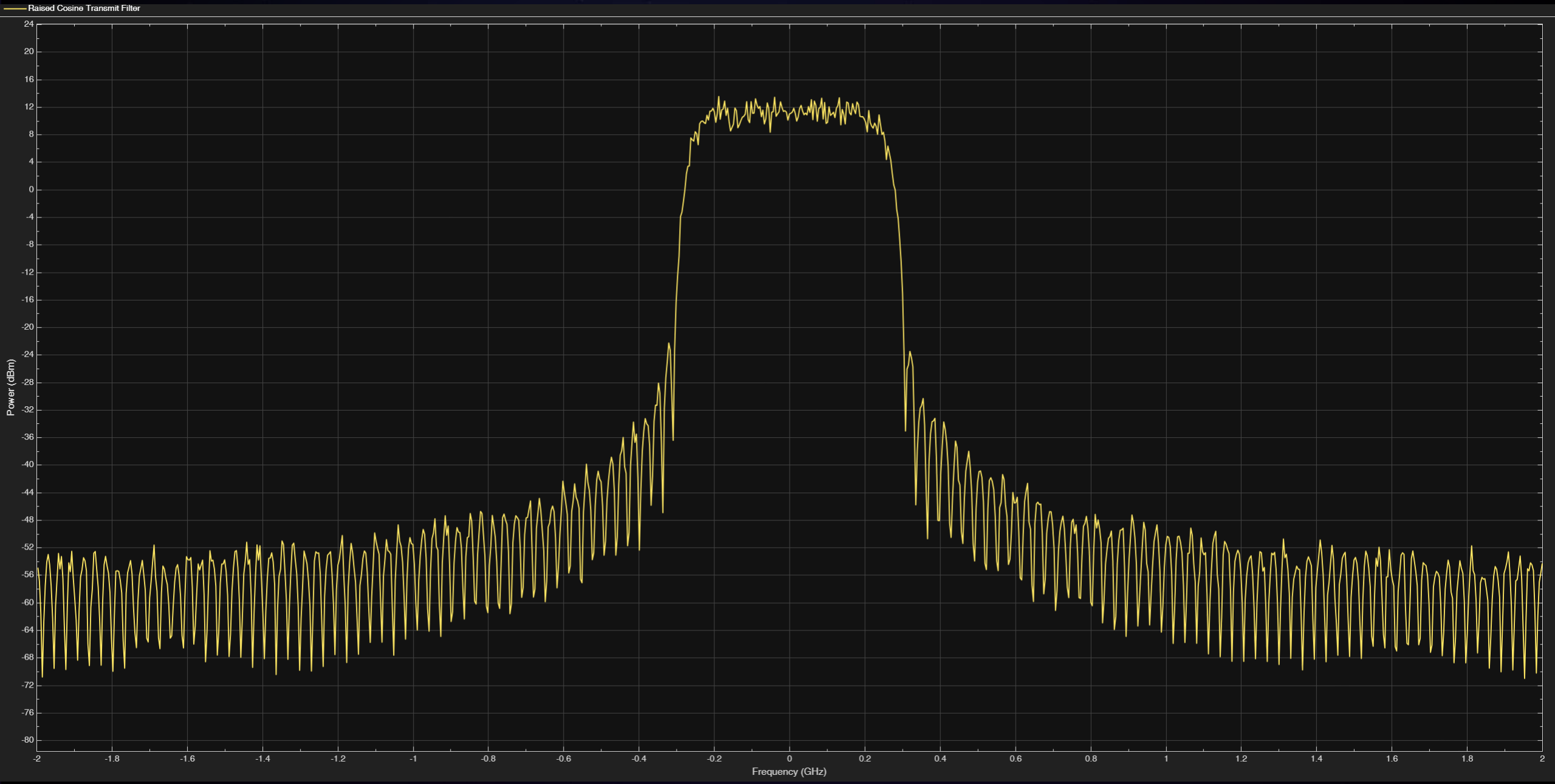
256 – APSK SPECTRUM AFTER UP CONVERSION



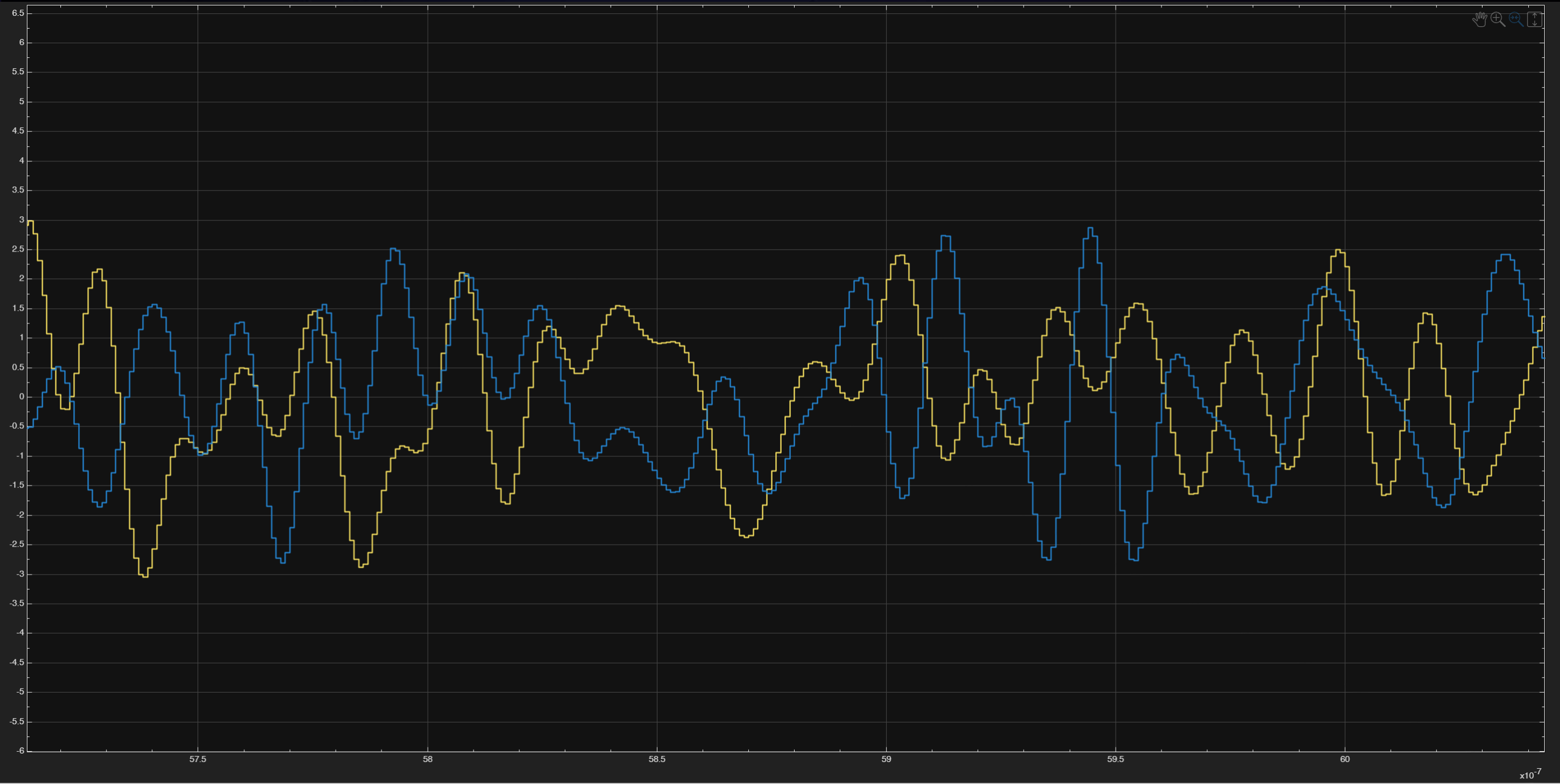
64 – APSK CONSTELLATION



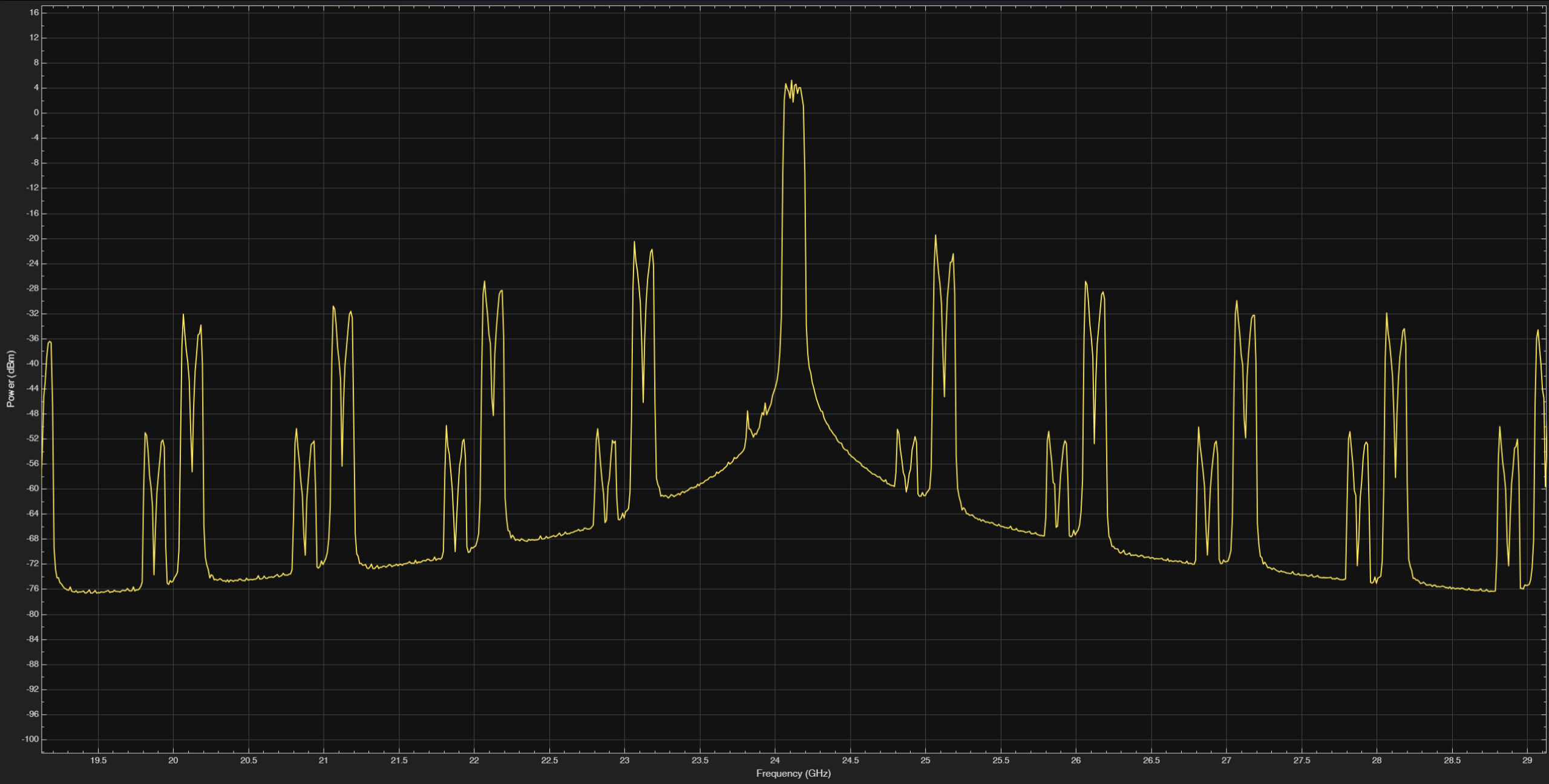
64 – APSK SPECTRUM



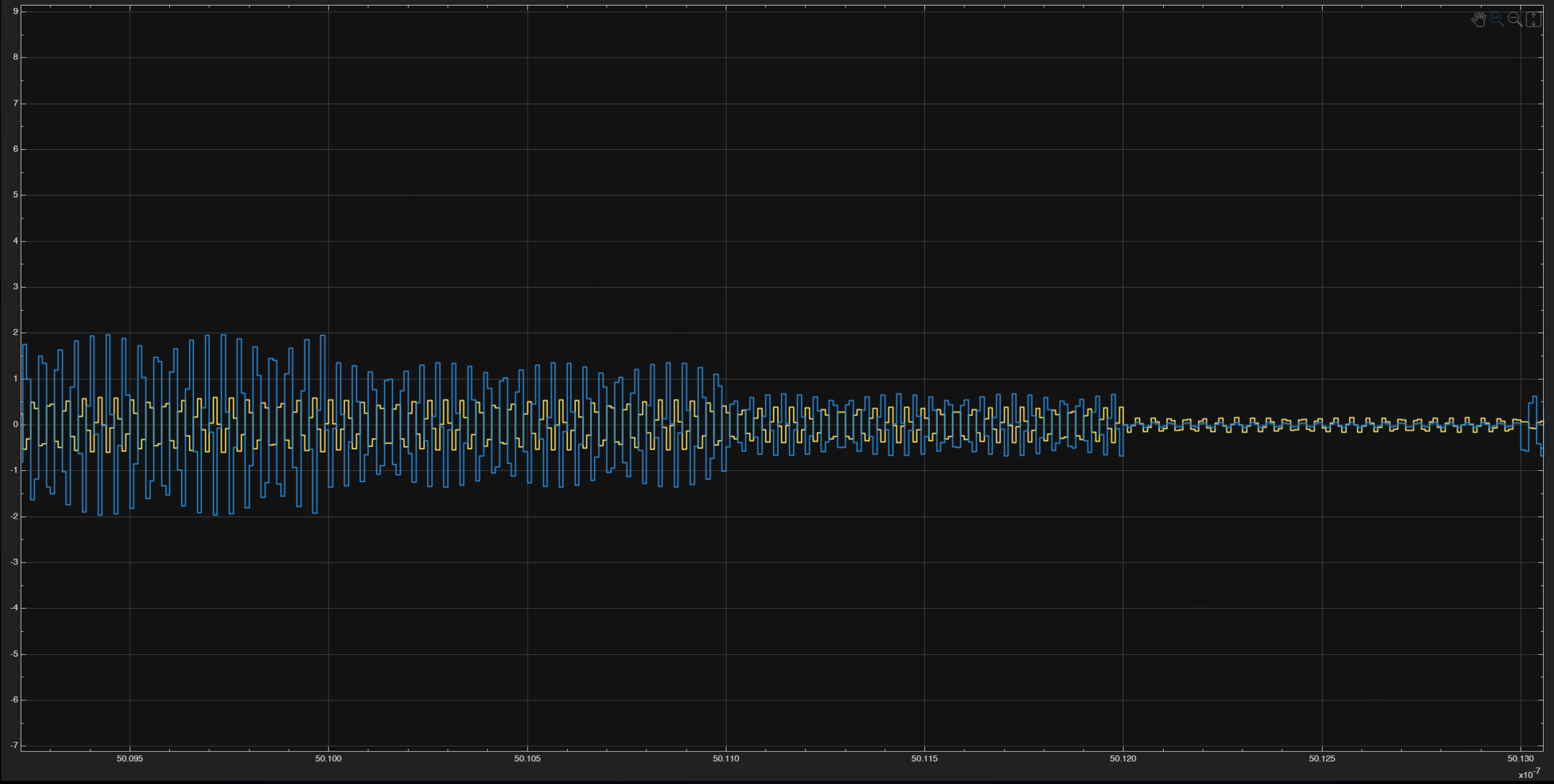
64 – APSK TIME DOMAIN



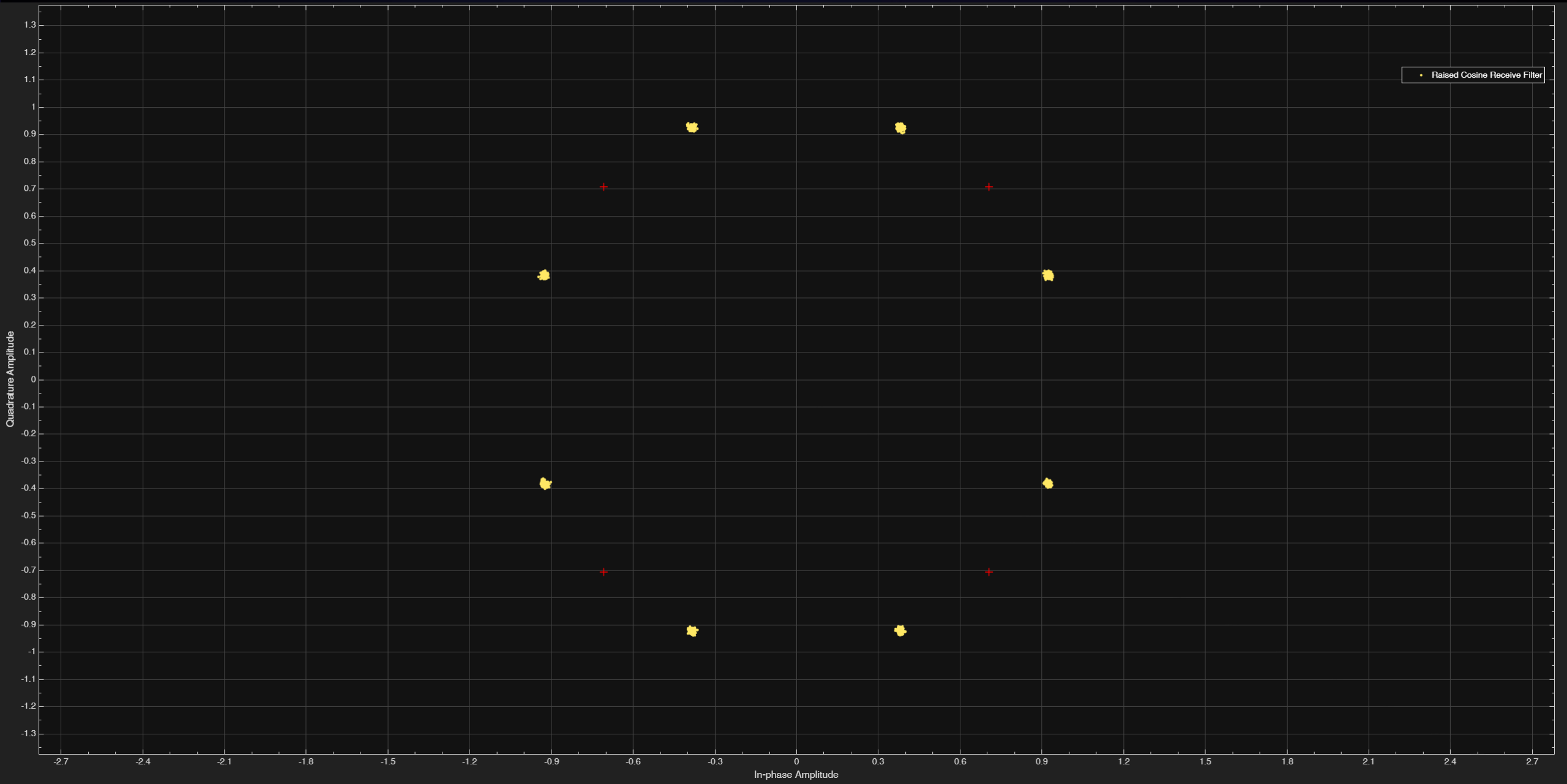
64 – APSK AFTER UP CONVERSION



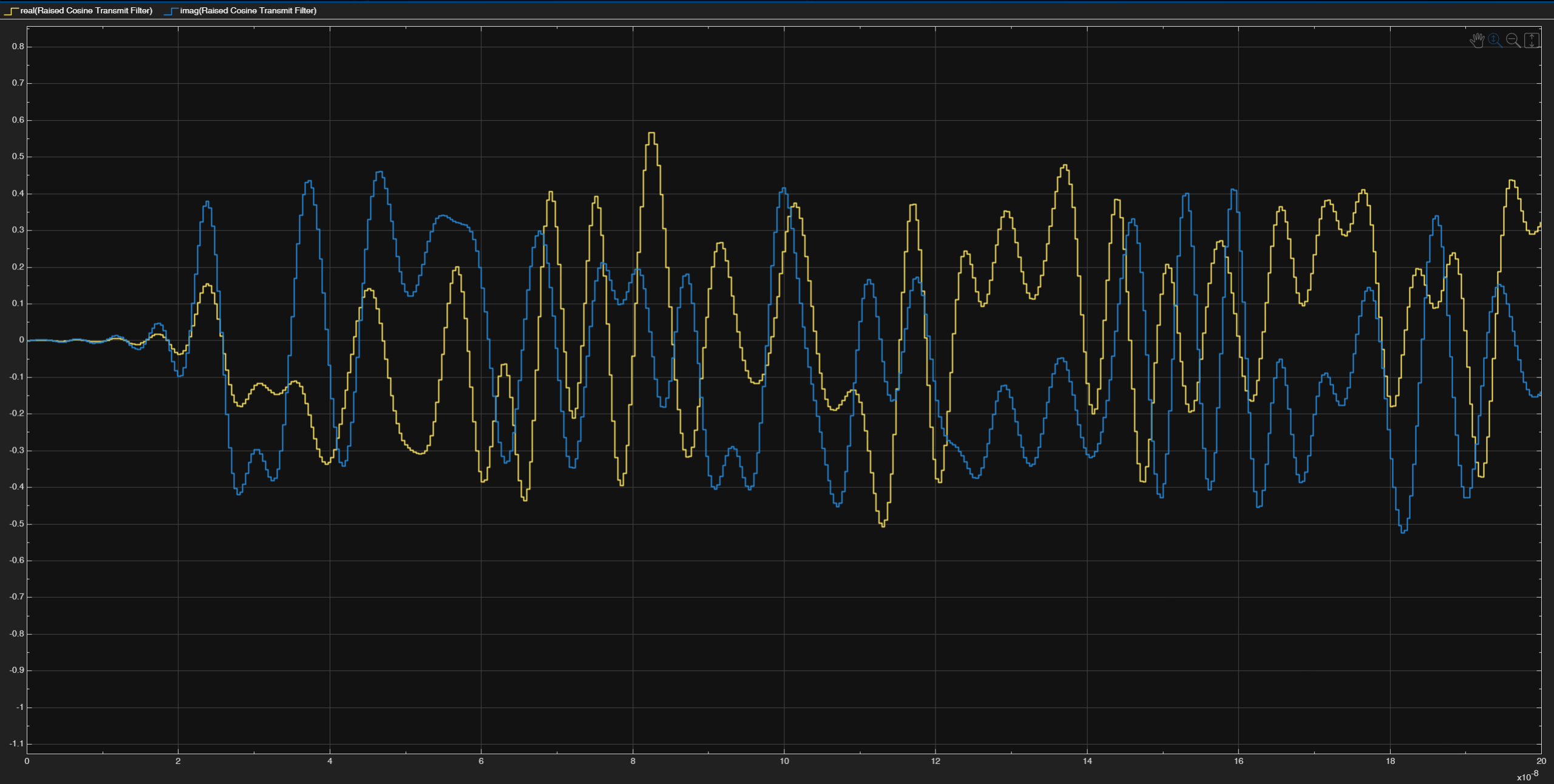
64 – APSK TIME DOMAIN AFTER UP CONVERSION



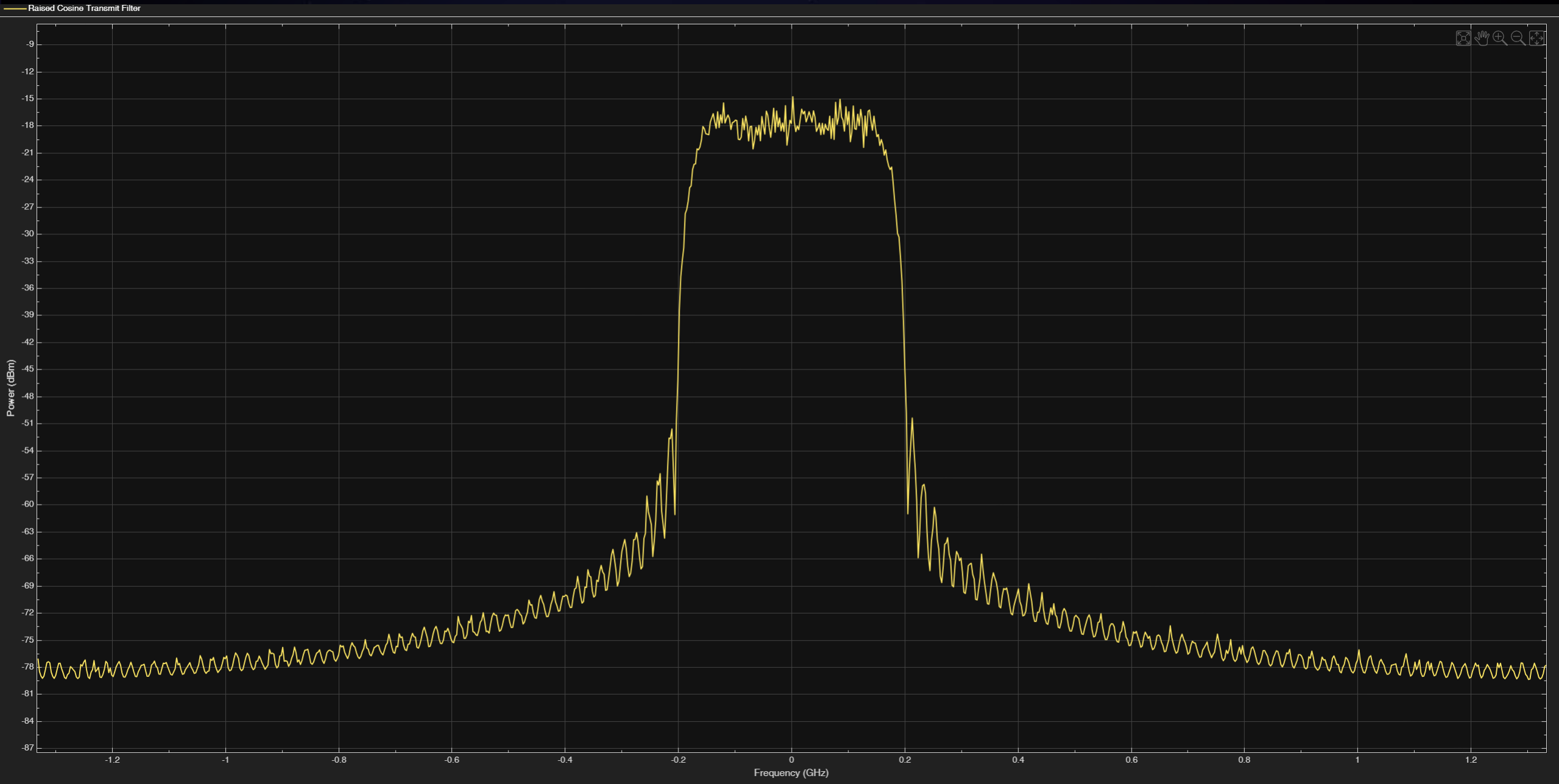
8 – PSK CONSTELLATION



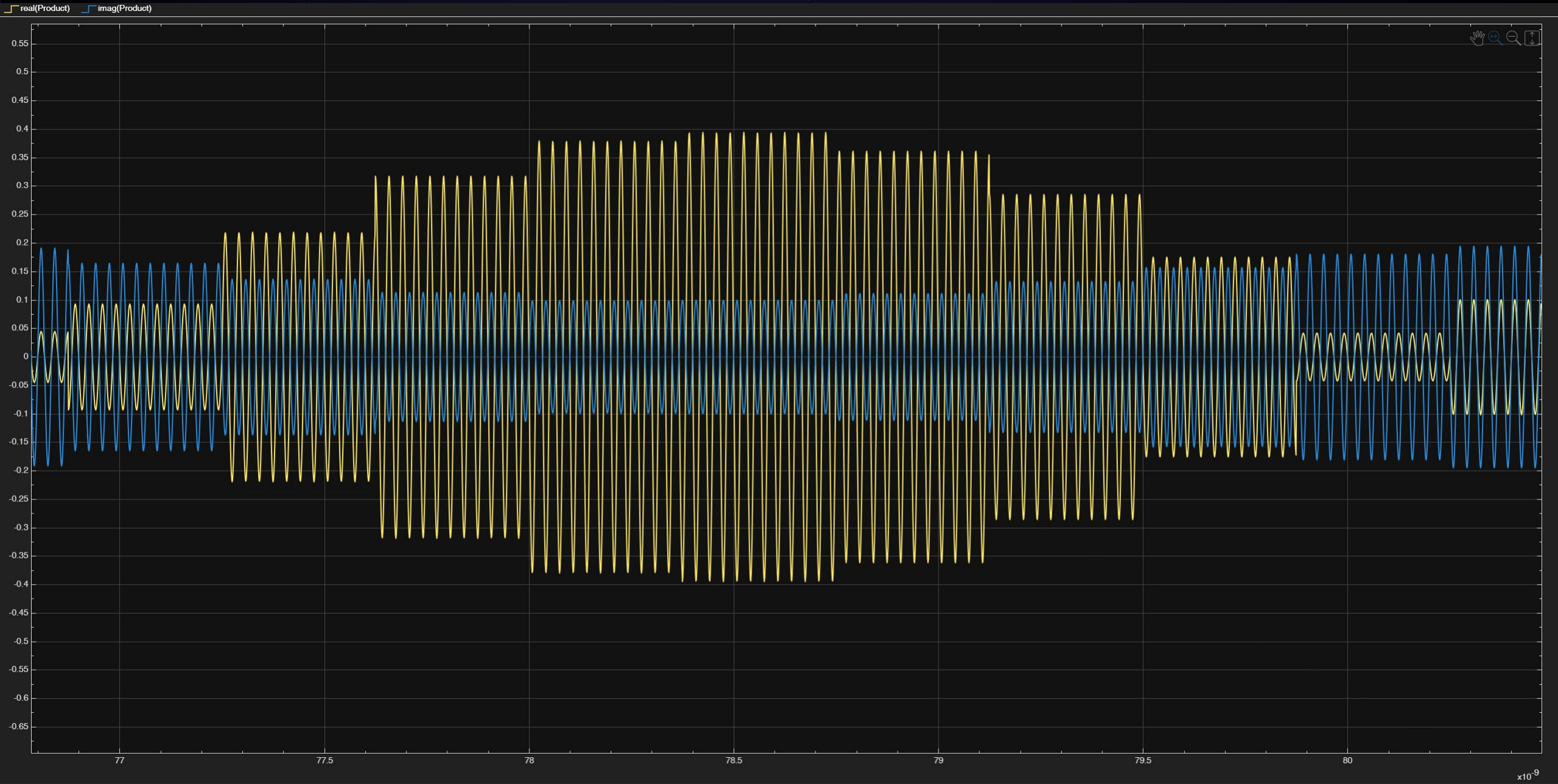
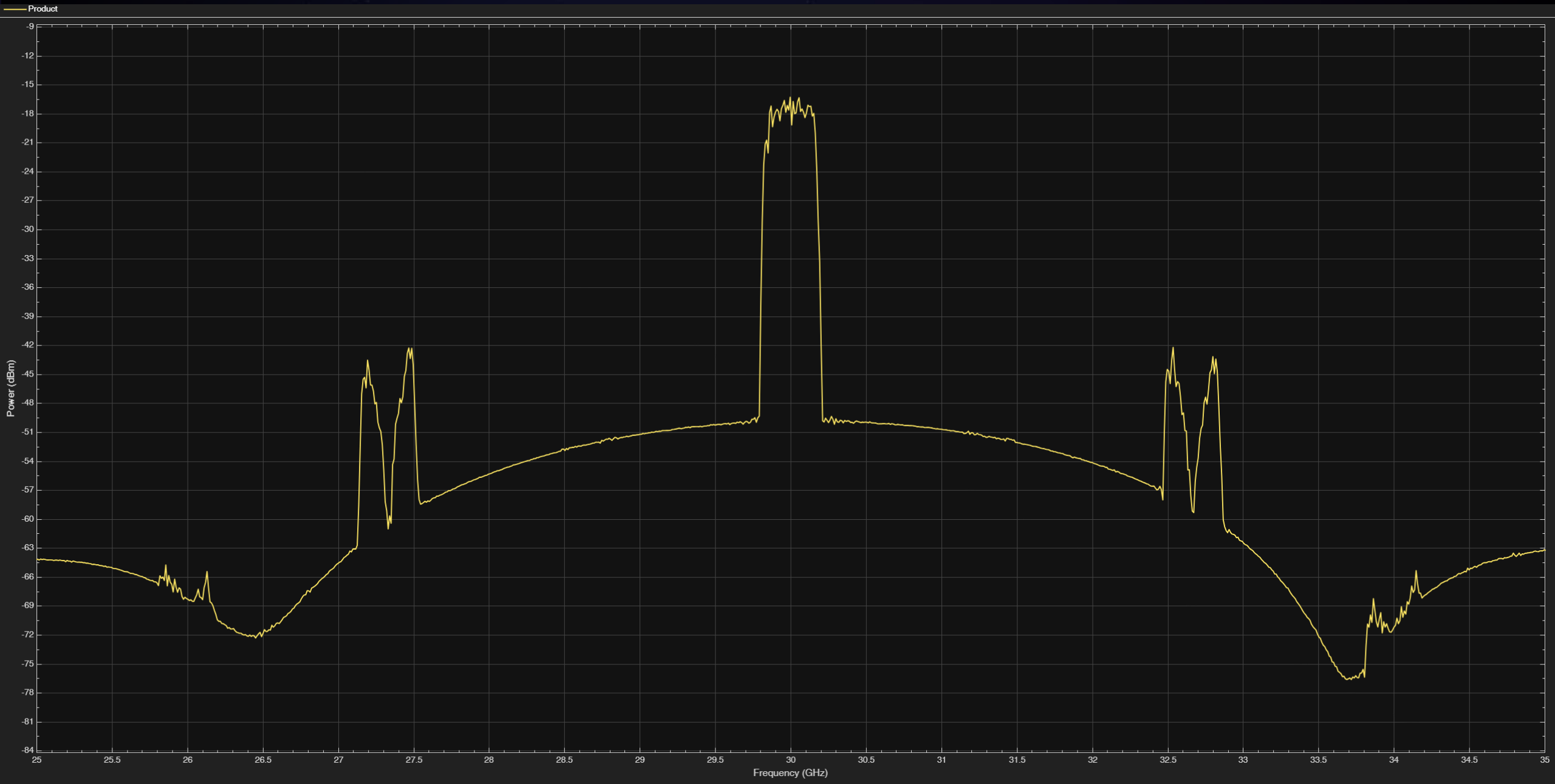
8 – PSK TIME DOMAIN



8 – PSK SPECTRUM



TIME DOMAIN AFTER UP CONVERSION



8 – PSK SPECTRUM AFTER UP CONVERSION